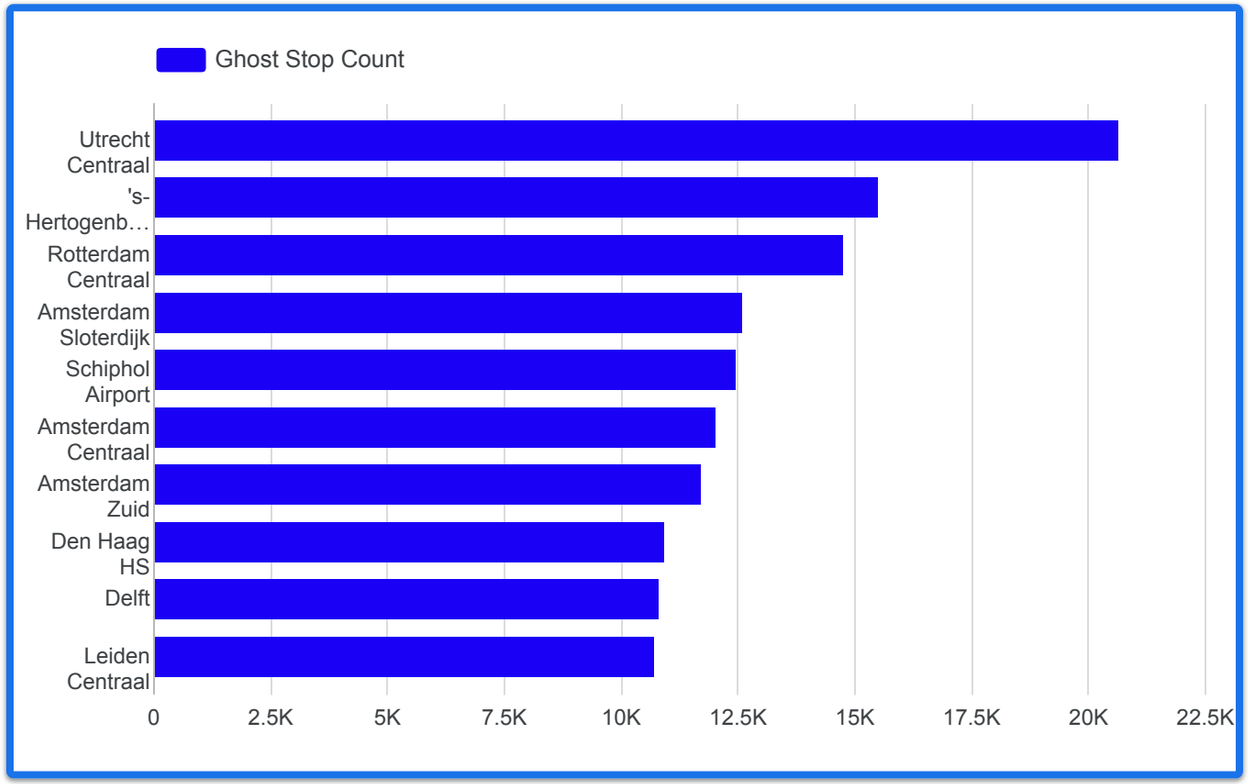
The Black Hole Audit: Top 10 Stations with Systemic Service Failures

Total Scheduled Stops	Total Systemic Failures	Affected Stations	Systemic Reliability Index (Target: < 0.01%)
22,226,311	748,221	523	0.03%

	Station Name	Ghost Stop Count
1.	Utrecht Centraal	20,649
2.	's-Hertogenbosch	15,509
3.	Rotterdam Centraal	14,761
4.	Amsterdam Sloterdijk	12,587
5.	Schiphol Airport	12,459
6.	Amsterdam Centraal	12,018
7.	Amsterdam Zuid	11,720
8.	Den Haag HS	10,933
9.	Delft	10,818
10.	Leiden Centraal	10,717
11.	Others	616,050
	Total	**748,221**



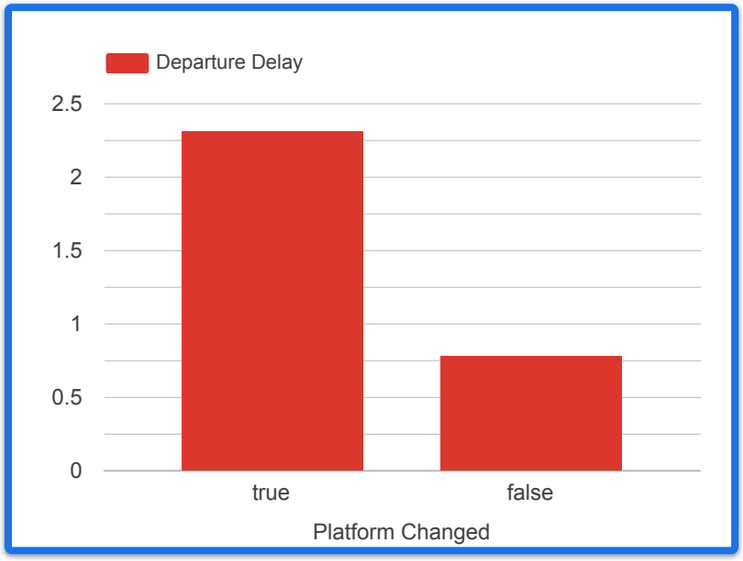
Pipeline Metadata :

Processed 22.2M rows in 13s via DuckDB + BigQuery BI Engine.
Data partitioned by service_date to optimize query costs.

The Punctuality Matrix: Peak-Hour Reliability by Operator

Arrival Hour / Arrival Delay

Service Company	1AM	2AM	12AM	11PM	6PM	7PM	5PM	10PM	8PM	4PM
211	-	0	-	11	3	2.5	-	16.75	16.25	-
Eu Sleeper	4.07	1.26	0	5.46	7.31	7.68	9.16	7.56	7.54	-
202	0	0.16	0.06	0.26	10.34	10.95	6.95	3.3	8.76	9.24
200	-	-	-	-	-	-	-	-	-	-
NS Int	4.87	4.47	4.25	5.44	6.7	6.64	6.14	5.7	6.68	5.35
TCS OV	-	0	-	-	4.54	3.9	5.64	3.67	2.91	4.33
TCS	-	-	-	-	0.39	2.06	2.22	-	1.81	0.24
VIAS	0	0	0	2.14	2.21	2.21	2.09	2.16	2.22	2.14
970	-	-	-	-	5.33	1.83	5.72	-	1.08	0.92
Eurobahn	-	0	-	0.73	2.13	2.21	2.16	1.73	1.92	2.25
NMBS	-	-	0	0.19	1.2	1.07	1.23	0.15	0.98	1.29
63	-	-	0.6	1.42	1.56	1.72	0.93	1.62	1.15	1.13
NS	1.21	0.82	1.18	1.11	1.01	0.99	0.99	1.05	0.95	0.94
154	-	-	1.04	0.59	1.61	1.05	1.86	0.71	1.2	1.39
RRReis K	1.43	-	0.94	0.97	0.81	0.72	0.88	0.67	0.59	0.85
137	-	-	0.16	0.29	1.99	0.85	1.85	0.13	0.58	1.64
Arriva	2.09	3.23	1.17	1.03	0.74	0.71	0.72	0.84	0.73	0.73
155	-	-	0.17	0.48	0.57	1.57	0.84	0.37	0.76	0.7
Blauwnet A	-	-	2.05	1.15	0.81	0.79	0.66	0.78	0.71	0.55
RRReis	0.65	-	0.91	0.98	0.77	0.64	0.76	0.71	0.62	0.65
Blauwnet	1.57	0	1.14	1.04	0.67	0.75	0.66	0.84	0.73	0.68
R-net	0	0	0.9	0.61	0.81	0.51	0.84	0.42	0.42	0.71

The Smoking Gun

Impact of Last-Minute Platform Changes on Departure Delays